

**UNIVERSITY OF SOUTHAMPTON**  
Faculty of Engineering and Physical Sciences  
School of Electronics and Computer Science

A project report submitted for the award of  
MEng Computer Science With Cybersecurity

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Examiner: ???

**A Reproducible Evaluation of  
Linux Rootkit Detectors via the  
Implementation of an Automated  
Testbed & Proof-of-Concept  
Rootkit**

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ABSTRACT

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This work is all about . . .



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$w$  The weight vector



## Acknowledgements

Thanks to no one.



*To ...*



# Chapter 1

## Background & Literature Review

### 1.1 Rootkits: Definition & Context

Rootkits are typically described as a specialised piece of malware built for persistence & stealth. They may additionally allow for functionality such as backdoors (cite magic bpf), file hiding or other abilities that give an attacker greater ability for data exfiltration and longevity on a system.

Linux has been chosen for this project due to its relevance and prevalence in cloud & server environments. Additionally, a wide range of rootkit techniques exist for Linux which will be covered in the following section. As a result many open-source tools exist in order to analyse a system for the presence of a rootkit. This can be a challenging task due to the very nature of a rootkit but is made additionally challenging when discussing **on-host** solutions.

### 1.2 Analysis of Previous Evaluation Methodologies

This is my literature review. Testing testing [Smith \(2024\)](#).



# Bibliography

Jane Smith. A paper title. *Journal Name*, 12:1–10, 2024.